



Mock for Quiz1



This assignment will not be graded and is only for practice.

If $A + 4I = 0A + 4I = 0$, where AA is a 3×3 matrix and II is an identity matrix of order 3, then find out the $\det(A)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -64

1 point

2 points

If $\{v_1, v_2, v_3\}$ forms a basis of R^3 , then which of the following are true?

☐

$\{v_1, v_2, v_1 + v_3\}$ forms a basis of R^3 .

☐

$\{v_1, v_1 + v_2, v_1 + v_3\}$ forms a basis of R^3 .

☐

$\{v_1, v_1 + v_2, v_1 - v_3\}$ forms a basis of R^3 .

☐

$\{v_1, v_1 - v_2, v_1 - v_3\}$ forms a basis of R^3 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{v_1, v_2, v_1 + v_3\}$ forms a basis of R^3 .

$\{v_1, v_1 + v_2, v_1 + v_3\}$ forms a basis of R^3 .

$\{v_1, v_1 + v_2, v_1 - v_3\}$ forms a basis of R^3 .

$\{v_1, v_1 - v_2, v_1 - v_3\}$ forms a basis of R^3 .

2 points

Let the set $\{(a, b), (c, d)\}$ be a linearly independent subset of R^2 . Choose the set of correct options.

☐

$\{(a, b, 0), (c, d, 0)\}$ must be a linearly independent subset of R^3 .

☐

$\{(a, b, 0), (c, d, 0), (0, 0, 1)\}$ must be a linearly independent subset of R^3 .

☐

$\{(a, b, 0), (c, d, 0), (1, 0, 0)\}$ must be a linearly independent subset of R^3 .

☐

$\{(a, b, 0), (c, d, 0), (0, 1, 0)\}$ must be a linearly independent subset of R^3 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(a, b, 0), (c, d, 0)\}$ must be a linearly independent subset of R^3 .

$\{(a, b, 0), (c, d, 0), (0, 0, 1)\}$ must be a linearly independent subset of R^3 .



Suppose $A = \begin{bmatrix} -3 & 5 & -1 \\ 0 & -2 & 3 \\ 7 & -4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 0 & x \\ x & -5 & 2 \\ -2 & 7 & 0 \end{bmatrix}$, where

$x \in \mathbb{N}$. If $\det(A) = \det(B)$, then find the value of x .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

2 points

Consider the following system of linear equations:

$$\begin{aligned} x - 3y + 2z &= 5 - a \\ 3x - 4y + z &= a + 1 \\ -3x + 9y - 6z &= 2a \end{aligned}$$

where $a \in \mathbb{R}$. If the given system of linear equations has infinitely many solutions, then what is the value of a ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 15

2 points

2 points

Consider the set $S = \{(1, -1, 0), (0, 1, 1), (2, 0, 1)\}$. Choose the correct set of options.

☐

$(7, -2, 2) \in \text{Span}(S)$

☐

$\text{Span}(S)$ is a proper subspace of $\mathbb{R}^3$.

☐

$\text{Span}(S) = \text{Span}\{(1, -1, 0), (0, 1, 1)\}$

☐

$\dim(\text{Span}(S)) = 3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(7, -2, 2) \in \text{Span}(S)$

$\dim(\text{Span}(S)) = 3$.

Suppose V is a vector space defined as $V = \{A \mid A \in M_{3 \times 3}(\mathbb{R}), \text{ the sum of the diagonal entries of } A \text{ is zero}\}$. What is the dimension of V ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

2 points

2 points

Consider the following sets with the given operations defined as follows:

$$V_1 = \{(a, b) \mid a, b \in R\}$$

Addition: $(a_1, b_1) + (a_2, b_2) = (a_1 + a_2, b_1 + b_2);$
 $(a_1, b_1), (a_2, b_2) \in V$
 Scalar multiplication: $\alpha(a, b) = (\alpha a, \alpha b);$
 $(a, b) \in V, \alpha \in R$

$$\{(a, b) \mid a, b \in R\}$$

Addition: $(a_1, b_1) + (a_2, b_2) = (a_1 + a_2, b_1 + b_2);$
 $(a_1, b_1), (a_2, b_2) \in V$ Scalar multiplication: $\alpha(a, b) = (\alpha a, \alpha b);$
 $(a, b) \in V, \alpha \in R$

$$V_2 = \{(a, b, c) \mid a, b, c \in R \text{ and } ab + bc = 0\}$$

Addition: $(a_1, b_1, c_1) + (a_2, b_2, c_2) = (a_1 + a_2, b_1 + b_2, c_1 + c_2);$
 $(a_1, b_1, c_1), (a_2, b_2, c_2) \in V$ V2 =
 Scalar multiplication: $\alpha(a, b, c) = (\alpha a, \alpha b, \alpha c);$
 $(a, b, c) \in V, \alpha \in R$

$$\{(a, b, c) \mid a, b, c \in R \text{ and } ab + bc = 0\}$$

Addition: $(a_1, b_1, c_1) + (a_2, b_2, c_2) = (a_1 + a_2, b_1 + b_2, c_1 + c_2);$
 $(a_1, b_1, c_1), (a_2, b_2, c_2) \in V$ Scalar multiplication: $\alpha(a, b, c) =$
 $(\alpha a, \alpha b, \alpha c);$
 $(a, b, c) \in V, \alpha \in R$

Choose the set of correct options.

☐

V_1 V1 is a vector space with respect the given operations.

☐

V_1 V1 is not a vector space with respect the given operations.

☐

V_2 V2 is a vector space with respect the given operations.

☐

V_2 V2 is not a vector space with respect the given operations.

No, the answer is incorrect.

Score: 0

Accepted Answers:

V_1 V1 is not a vector space with respect the given operations.

V_2 V2 is not a vector space with respect the given operations.

2 points

Find out the correct statements from the following:

☐

The set of all invertible matrices of order 22 forms a subspace of $M_2(R)$ $M_2(R)$ with respect to usual addition and scalar multiplication.

☐

The vector space R^5 R^5 (with respect to usual additional and scalar multiplication) contains two disjoint subspaces.

☐

There exist three linearly dependent vectors in R^3 R^3 such that none of the three is a multiple of another.

☐

Union of two linearly independent sets need not be linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

There exist three linearly dependent vectors in R^3 R^3 such that none of the three is a multiple of another.

Union of two linearly independent sets need not be linearly independent.

2 points

If S_1 is a maximal linear independent set and S_2 is a minimal spanning set of a vector space V , then which of the following option(s) is (are) true?

☐

For any $v \in S_1$, $S_1 \setminus \{v\}$ is a linearly independent.

☐

For any $v \in S_2$, $S_2 \setminus \{v\}$ is a spanning set of V .

☐

For any $v \in V \setminus S_1$, $S_1 \cup \{v\}$ is a linearly dependent.

☐

For any $v \in V$, $S_2 \cup \{v\}$ is a spanning set of V .

No, the answer is incorrect.

Score: 0

Accepted Answers:

For any $v \in S_1$, $S_1 \setminus \{v\}$ is a linearly independent.

For any $v \in V \setminus S_1$, $S_1 \cup \{v\}$ is a linearly dependent.

For any $v \in V$, $S_2 \cup \{v\}$ is a spanning set of V .

1 point

Consider the following statements:

Statement 1: $0(0, 0) + 0(1, 1) = (0, 0)$ implies that the set $\{(0, 0), (1, 1)\}$ is a linearly independent subset of R^2 .

Statement 2: $2(1, 0) + 2(0, 1) = (2, 2)$ implies that the set $\{(1, 0), (0, 1), (2, 2)\}$ is a linearly dependent subset of R^2 .

Which of the following options is correct?

- ☐ Both the statements are true.
☐ Statement 1 is true, but Statement 2 is false.
☐ Statement 2 is true, but Statement 1 is false.
☐ Both the statements are false.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Statement 2 is true, but Statement 1 is false.

Three people denoted by P_1, P_2, P_3 intend to buy some rolls, buns and cakes from a shop. They pay ₹ x_1 , ₹ x_2 , ₹ x_3 per unit for rolls, buns and cakes, respectively. P_1 bought 11 unit of rolls, 22 units of buns, and 33 units of cakes, P_2 bought 22 units of rolls, 44 units of buns, and 55 units of cakes and P_3 bought 11 unit of rolls, 33 units of buns, and 44 units of cakes. The total amount spent by P_1, P_2, P_3 are ₹ 1616, ₹ 2929, ₹ 2121, respectively. Answer the following questions using this given data.

Represent the system of linear equations above in terms of its matrix form $Ax = b$, where

$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} 1616 \\ 2929 \\ 2121 \end{bmatrix}$. Find the $(2, 2)$ -th cofactor of A ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

If $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ is the solution of $Ax = b$, then the value of $x_1 + x_2 + x_3$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

2 points

Let $(M_{m \times n}(R))$ denote the set of all $(m \times n)$ matrices with real entries and let (D) be a non-zero matrix such that $(D)(D)$ is the product of the matrices $(A \in M_{7 \times 12}(R), B \in M_{12 \times 3}(R))$ and $(C \in M_{3 \times 5}(R))$.

Find the maximum possible rank of $(D)(D)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Find the minimum possible rank of $(D)(D)$